DSAA3073: Theories in Data Science

Week 9 Validation Sheet · Expert Advice Under an Information Boundary

Coverage: Week 9 **Duration:** 30 minutes · **Total:** 130 marks

Answer all questions. For multiple-choice questions, select exactly one answer. Show the essential reasoning requested; unsupported answers may receive no credit.

Part A: Multiple choice

A1 [5 marks]. Which order is legal in the full-information online protocol?

(a) Reveal ℓ_t, choose p_t, incur loss, update.
(b) Choose p_t, reveal ℓ_t, incur loss, update.
(c) Choose p_t, update, reveal ℓ_t, incur loss.
(d) Reveal ℓ_t, update, then choose p_t.

A2 [5 marks]. External regret compares the learner with

(a) the best expert chosen separately on every round.
(b) the best fixed expert on the same realized loss sequence.
(c) the expert having the smallest population risk.
(d) the learner's own loss on an independent test sequence.

A3 [5 marks]. Which assumption is essential for the usual logarithmic Halving mistake bound?

(a) Every expert has positive weight.
(b) At least one expert is perfect on the complete sequence.
(c) Losses lie anywhere in $[-1, 1]$.
(d) The learner samples an expert privately.

A4 [5 marks]. Which formula is the course-primary **linear** Multiplicative Weights Update?

(a) $w_{t+1}(i) = w_{t(i)}\left(1 - \eta\ell_{t(i)}\right)$.
(b) $w_{t+1}(i) = w_{t(i)}e^{-\eta\ell_{t(i)}}$.
(c) $w_{t+1}(i) = 0$ whenever expert i makes one mistake.
(d) $x_{t+1} = x_t - \eta g_t$.

Part B: Definition in use

B1 [20 marks]. For each scenario, select the most appropriate **core Week 9 algorithm** and state the type of guarantee that its assumptions actually license. Give one sentence of justification per row.

1. Binary experts; a perfect expert is guaranteed on the complete sequence.
2. Binary experts; no perfect expert is promised; the learner must predict deterministically and must remain defined after every expert errs.
3. Binary experts; private sampling is allowed; the current loss is fixed without observing the private current sample.
4. General losses in $[0, 1]$; the complete loss vector is revealed only after commitment; no perfect expert is promised.

Answer space

Part C: Basic skills

C1 [7 marks]. On four rounds, three experts have loss sequences $h_1 : (0, 1, 1, 0)$, $h_2 : (1, 0, 0, 1)$, and $h_3 : (0, 0, 1, 1)$. The learner's loss sequence is $(1, 0, 1, 1)$. Compute the external regret against one best fixed expert on this same sequence. Also state why taking the lowest expert loss separately on each round would use the wrong comparator.

Answer space

C2 [8 marks]. Linear MWU has current weights $(2, 1, 1)$, learning rate $\eta = \frac{1}{4}$, and current loss vector $\ell_t = \left(1, 0, -\frac{1}{2}\right)$. Compute p_t, the mixture loss $p_t \cdot \ell_t$, all three multiplicative factors, and the updated weights.

Answer space

Part D: Construction and counterexample

D1 [17 marks]. Three binary experts make the following predictions on two rounds:

round	h_1	h_2	h_3
1	0	0	1
2	0	1	1

Halving starts with all experts active, uses majority vote, and the labels are $y_1 = 1$, $y_2 = 0$. Trace the active set after each round. Compute the best fixed expert's two-round loss, and use the trace to explain exactly which line of the logarithmic Halving proof fails. Why does the mere existence of a best expert not repair the proof?

Answer space

D2 [18 marks]. On two rounds, a randomized learner uses the same distribution $p_t = \left(\frac{3}{4}, \frac{1}{4}\right)$ and the revealed loss vector is $\ell_t = (1, 0)$ on both rounds. The loss vectors are fixed without observing either private current sample.

Compute the cumulative mixture loss and the expected cumulative sampled loss. Then construct a possible two-round sample path that disproves the claim "the sampled cumulative loss always equals (or is at most) the cumulative mixture loss." State the information condition and the precise guarantee that remain valid.

Answer space

Part E: Proof and derivation

E1 [25 marks]. There are n experts and losses $\ell_{t(i)} \in [-1, 1]$. Before ℓ_t is revealed, a developer chooses a predictable step size $0 < \eta_t \le \frac{1}{2}$, forms $p_{t(i)} = \frac{w_{t(i)}}{\sum_j} w_{t(j)}$, and commits using p_t. After the full loss vector is revealed, the weights are updated by

$$w_{t+1}(i) = w_{t(i)}\Big(1 - \eta_t \ell_{t(i)}\Big), \quad w_1(i) = 1.$$

The developer claims that allowing any such predictable sequence (η_t) automatically preserves the ordinary unweighted sublinear external-regret guarantee of constant-rate linear MWU.

Audit this claim as one integrated argument. Derive the strongest fixed-comparator certificate that follows from the stated update while retaining the comparator terms $\eta_t^2 \left|\ell_{t(i)}\right|$. Explain the weighting or quantifier error in reading that certificate as ordinary external regret. Then construct a concrete legal step-size schedule and a two-expert loss sequence for which the ordinary external regret is linear. Finally, state what additional step-size or tuning structure recovers the constant-rate Week 9 guarantee. You may use $1 - y \le e^{-y}$ for all real y and $\ln(1 - \eta x) \ge -\eta x - \eta^2 |x|$ when $|x| \le 1$ and $0 < \eta \le \frac{1}{2}$.

Answer space

Part F: Synthesis and limitation

F1 [15 marks]. Complete a comparison of the three core algorithms below. For each one, identify (i) what is updated or sampled, (ii) the update shape, and (iii) the strongest guarantee type licensed by its stated setting. Your comparison must distinguish deterministic from expected/pathwise claims.

- deterministic Weighted Majority for binary mistakes;
- Randomized Weighted Majority for binary mistakes;
- linear MWU for bounded full-information losses.

A colleague also writes $w_{t+1}(i) = w_{t(i)}e^{-\eta \ell_{t(i)}}$ and calls it "the same linear update." State the only conclusion about that formula that is licensed by the Week 9 core.

Answer space